

Autonomous Sorting Bookshelf

Problem Statement:



The system aims to **automatically** manage the books on a bookshelf, including book **shelving** and **retrieval** and borrow/returned status checks.

- Missions:**
- Handle book of various size
 - Fast and accurate actuation
 - Recognize books (e.g QR code)
 - Shelf and retrieve books
 - Record book location and thickness

Customer Needs & Specifications:

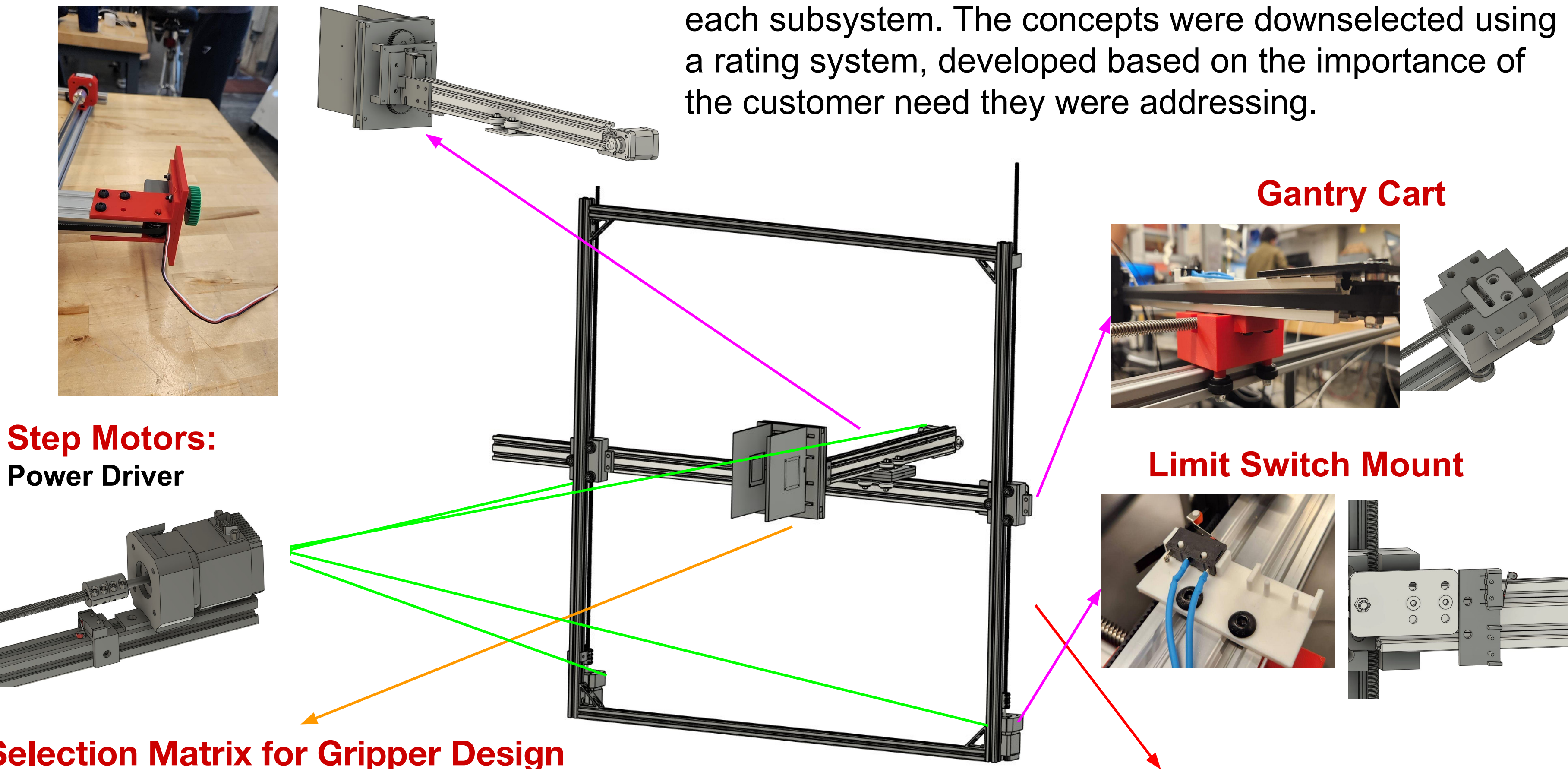
The gantry system must ensure precision within 2mm, efficient book sorting for quick location identification, and safety through fuss and cable chain systems with a 2-amp limit, addressing the customers' needs for accuracy, efficiency, and hazard-free operation.

Needs	Specification	Value
The system must ensure precise and reliable selection of the correct book.	Positioning precision must be maintained within a tolerance to ensure correct book selection.	< 2 mm
The system should streamline the process of finding books.	Must accurately extract metadata and book location information to enhance efficiency and eliminate unnecessary delays in the retrieval process.	Accurate information extraction & book location
The system must prioritize user and operational safety.	Safety features include a limitation for cables, providing effective electrical hazard protection and ensuring durability and safety of the mechanism.	2A limitation Cable protection

Concept Generation and Selection:

XYZ Linear Actuators:
Control gantry's XYZ movement

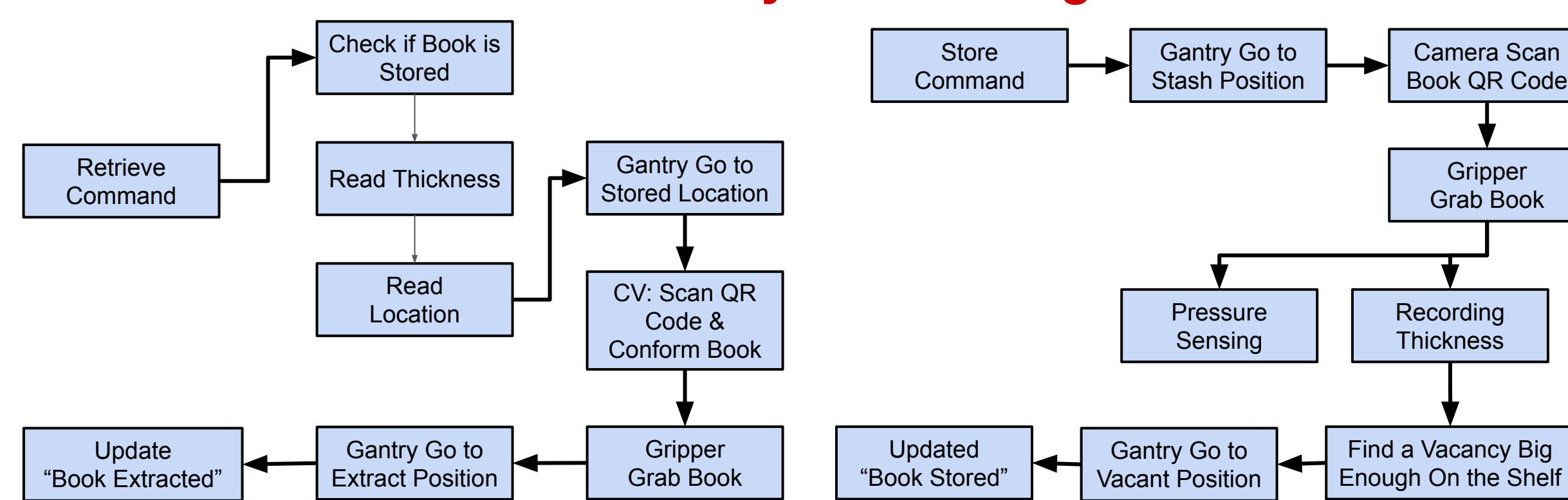
Our concepts were developed from our functional decomposition. Many design ideas were explored for each subsystem. The concepts were downselected using a rating system, developed based on the importance of the customer need they were addressing.



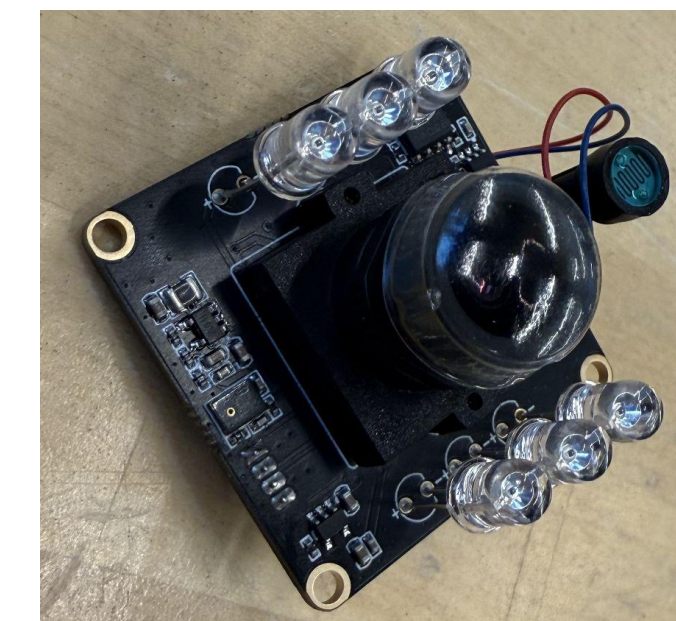
Selection Matrix for Gripper Design

Description	Claw	Gripper 1 (linkage)	Gripper 2 (rack & pinion)	Gripper 3 (rack & pinion)
MODEL				
Criteria	Weight	Datum	Datum	Datum
Stability	3	-	+	++
Maximum Load	3	--	++	++
Easy to Build	2	++	+	0
Cost	1	0	+	-
Robustness	2	-	+	+
Durable	2	-	0	-
+	4	14	11	14
0	1	2	4	0
-	13	1	1	7
Net Score	-9	13	10	7

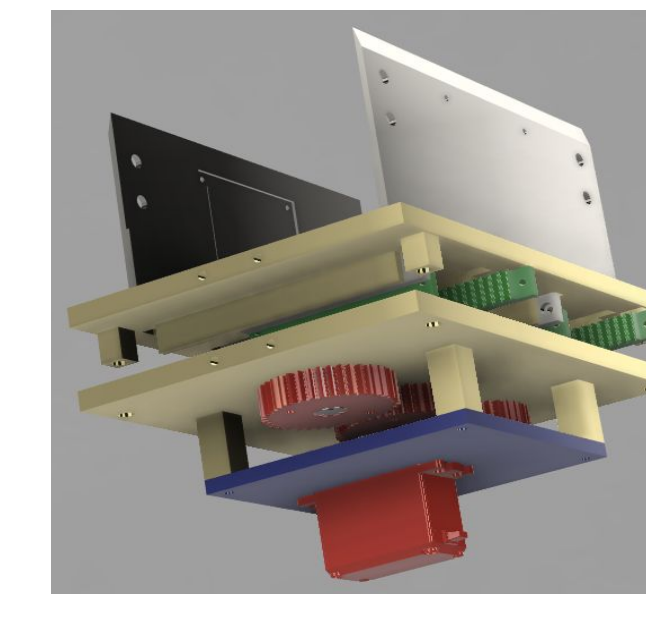
Gantry Control Logic



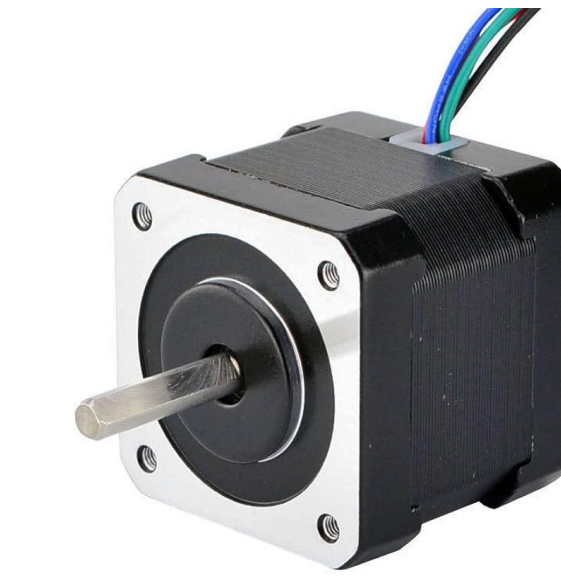
Concept Description:



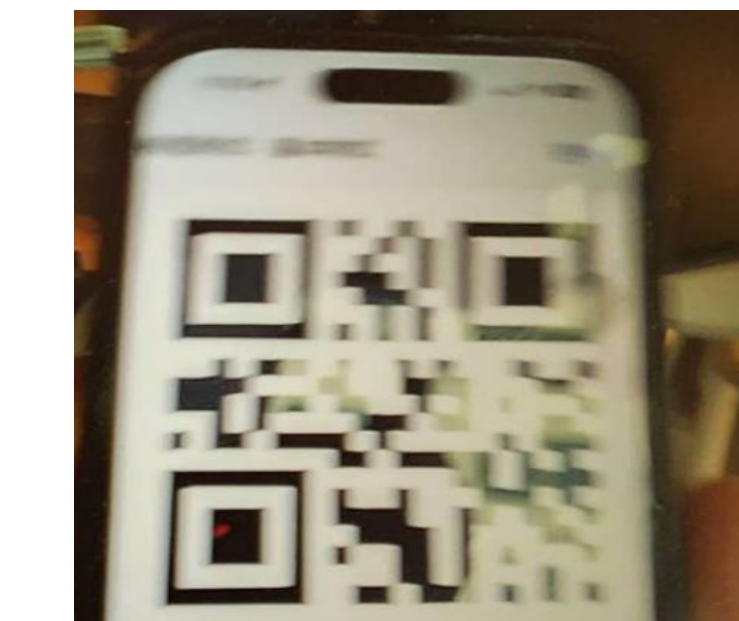
Camera Module (B0205)



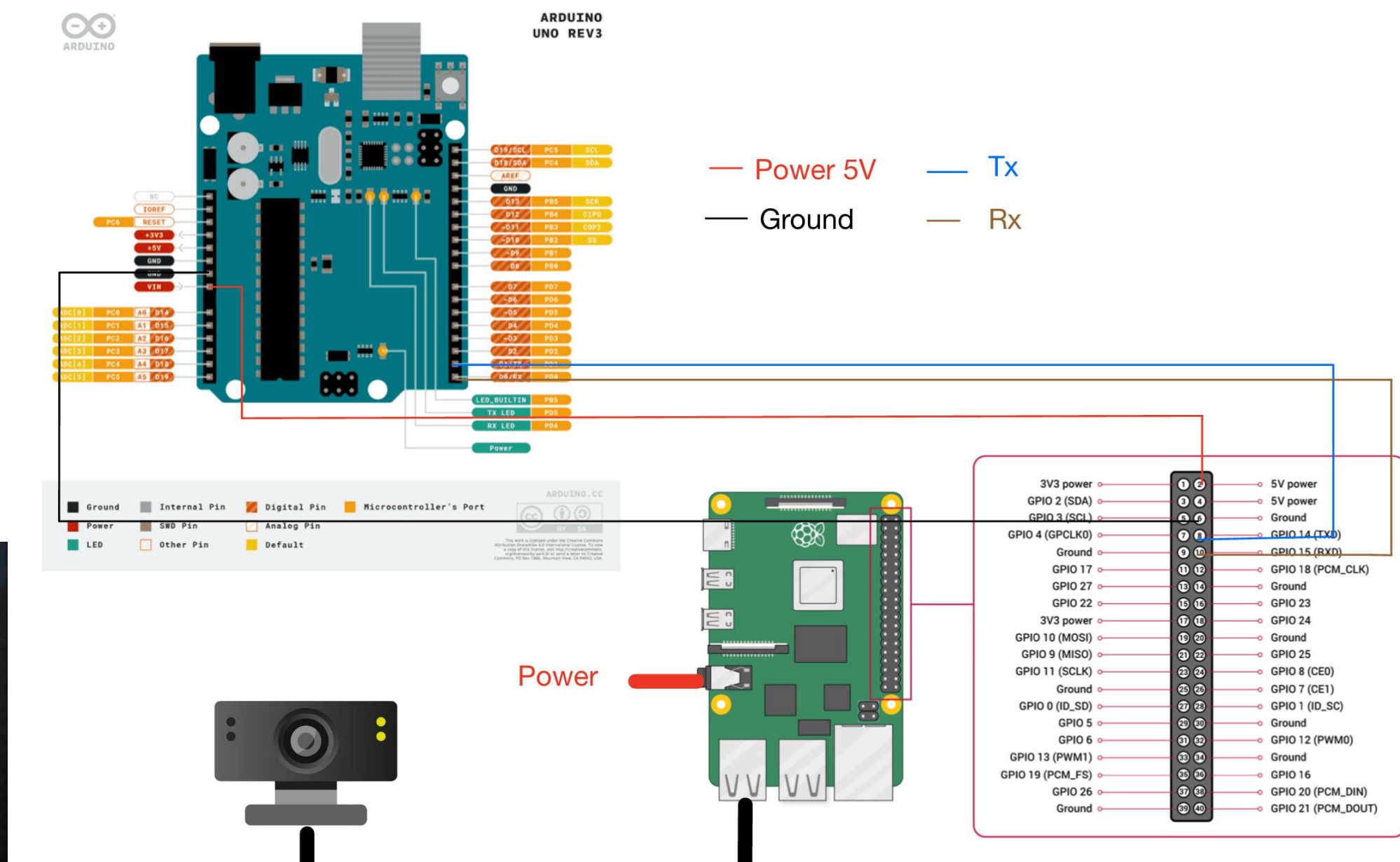
Gripper Design (Rack & Pinion)



NEMA 17 Motor



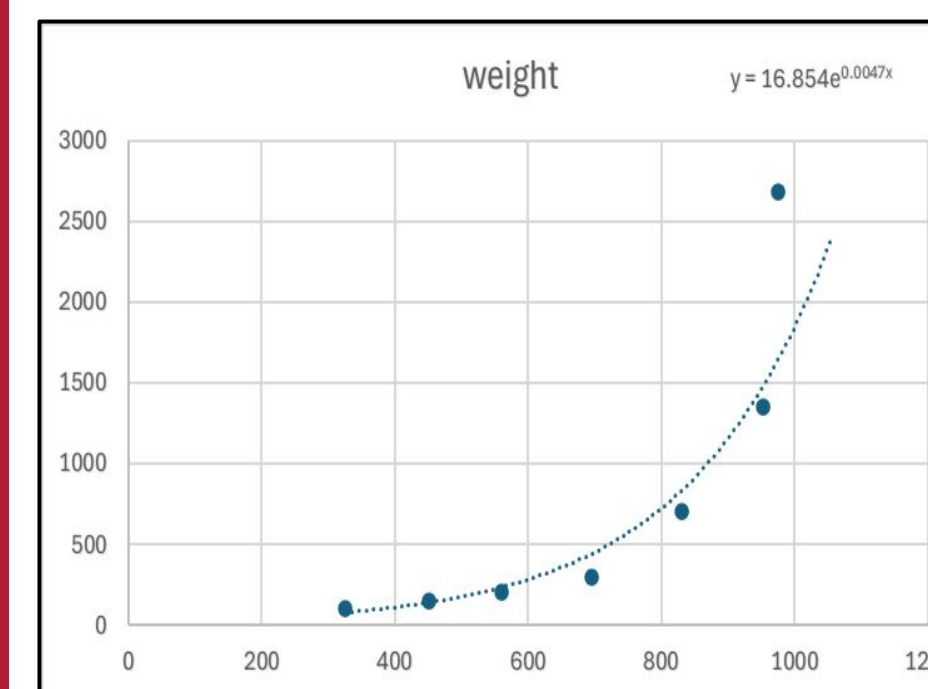
QR Code Scanner



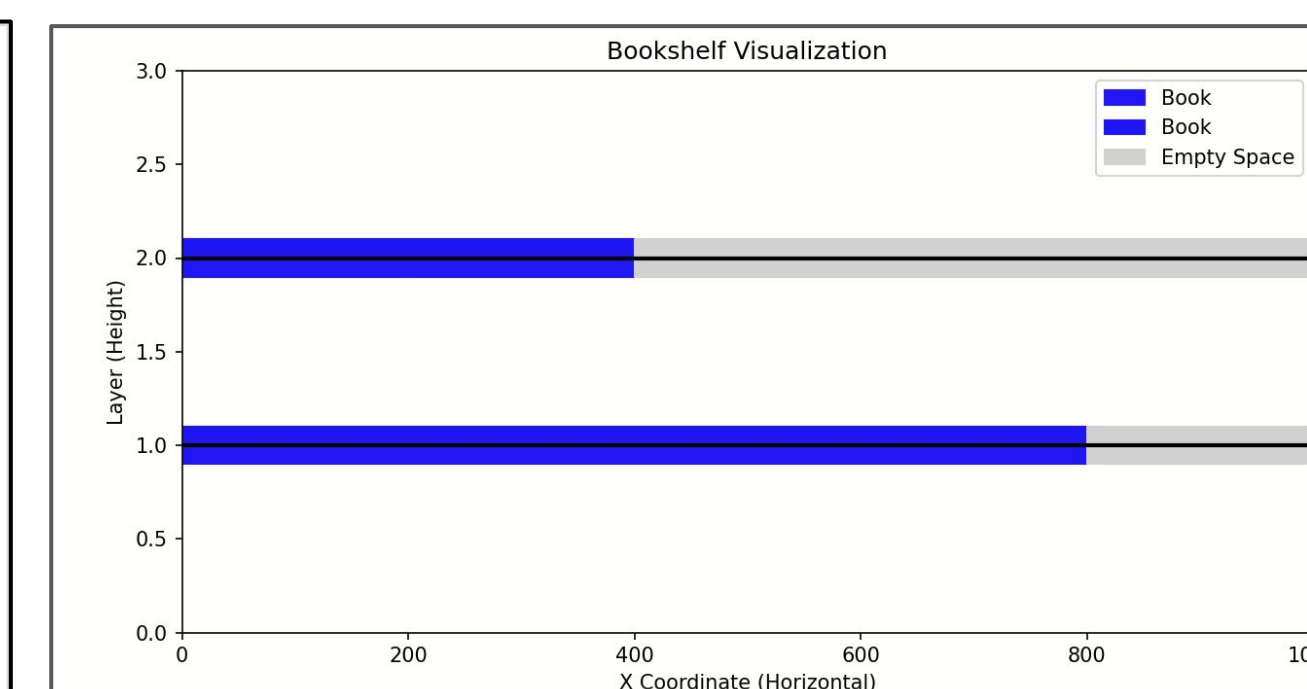
Circuit & Wiring Graph

Testing & Result:

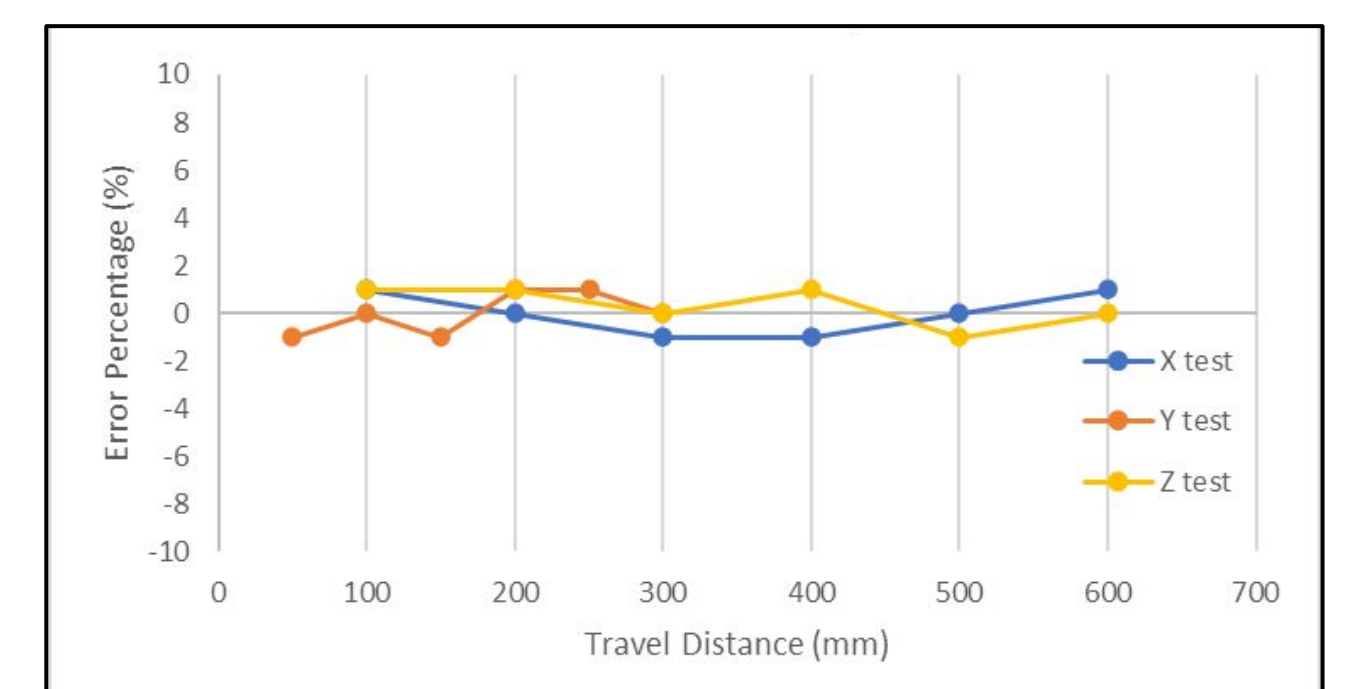
Force Reaction Sensor Test



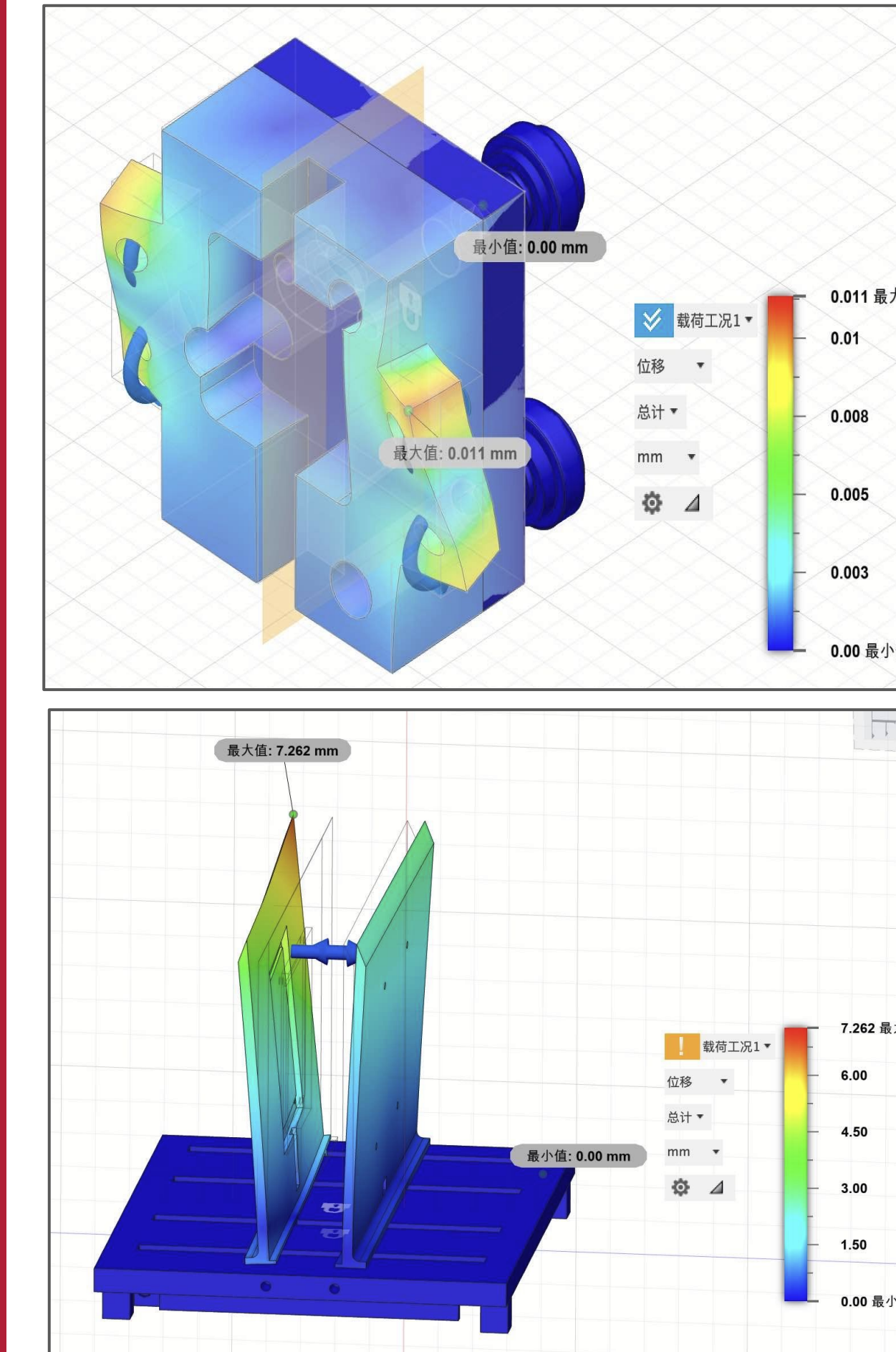
Bookshelf Status Simulation



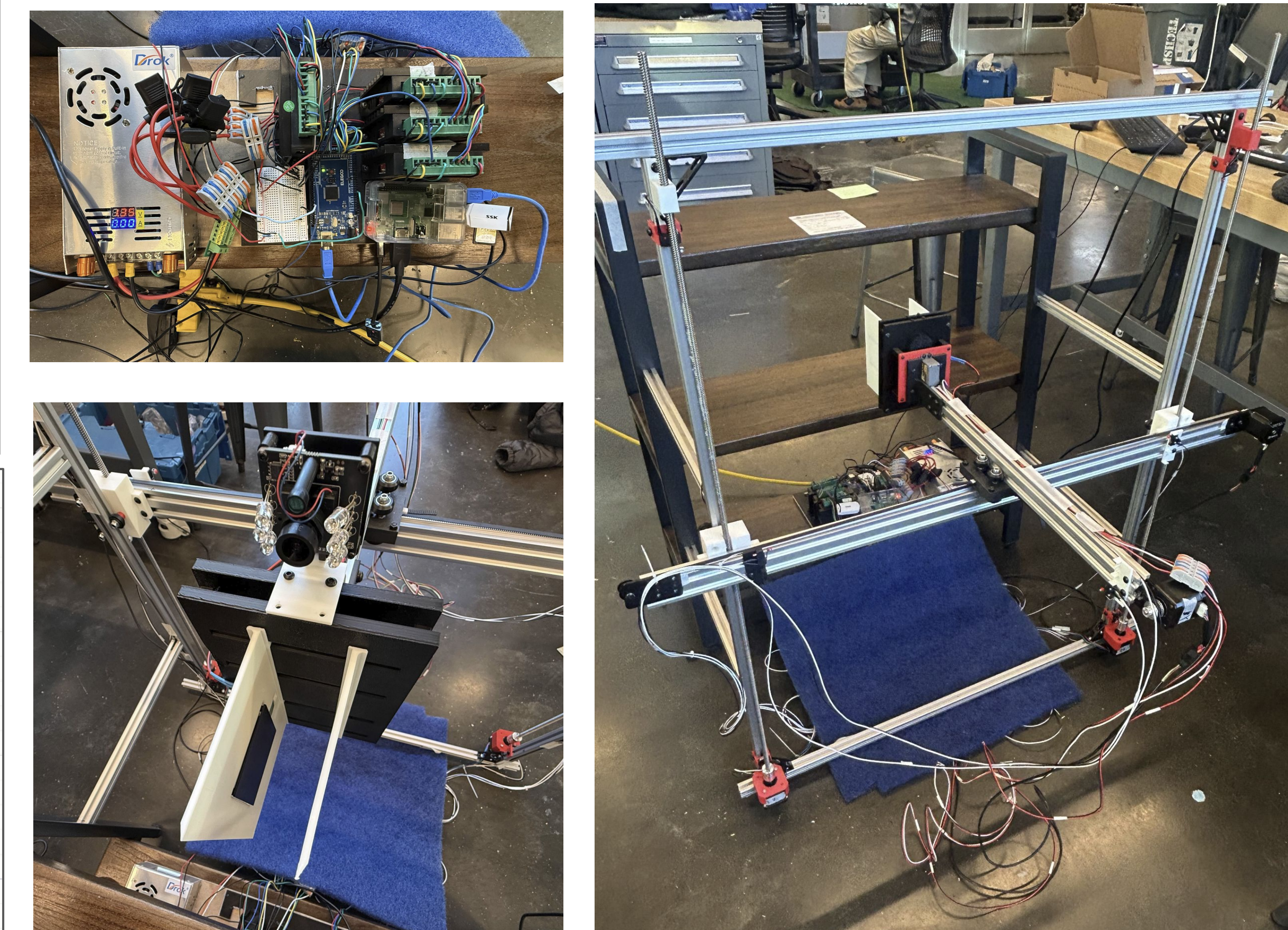
Actuator Accuracy



FEA Analysis of Critical Parts



Designed Prototype



Conclusion & Future Improvement

Core Deliverables:

- Accurately grip and retrieve books.
- Record and store book name and thickness.
- Locate books with precision.
- Grip books of various sizes and thicknesses stably.

Future improvements:

- Implement robust control, including error handling for obstacles or books removed manually.
- Enhance the user interface.